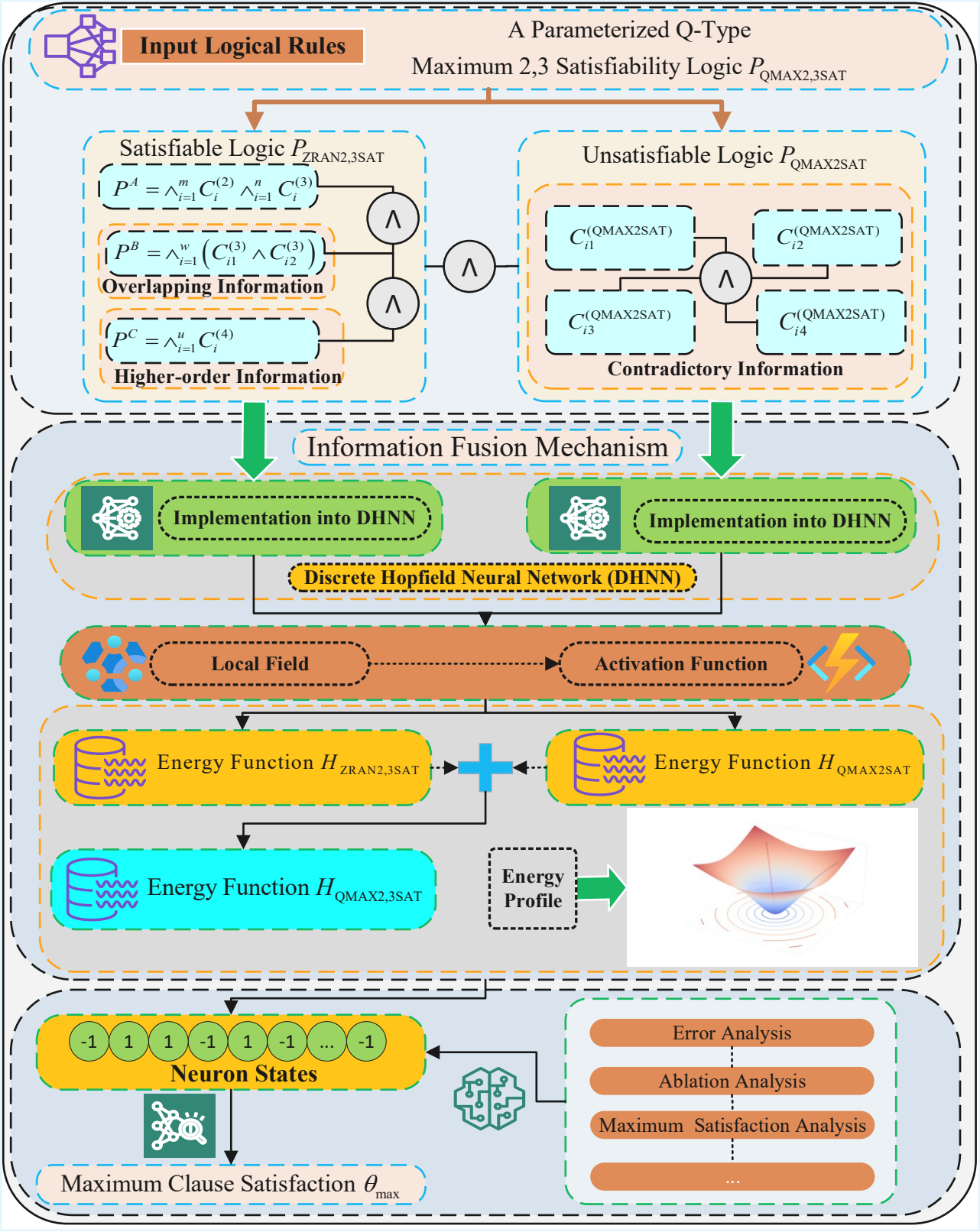
Input Logical Rules
A Parameterized Q-Type
Maximum 2,3 Satisfiability Logic $P_{\mathrm{QMAX2,3SAT}}$
Satisfiable Logic $P_{\mathrm{ZRAN2,3SAT}}$
$P^A = \wedge_{i=1}^{m} C_i^{(2)} \wedge_{i=1}^{n} C_i^{(3)}$
$P^B = \wedge_{i=1}^{w} \left(C_{i1}^{(3)} \wedge C_{i2}^{(3)} \right)$
Overlapping Information
$P^C = \wedge_{i=1}^{u} C_i^{(4)}$
Higher-order Information
Unsatisfiable Logic P_{QMAX2SAT}
$C_{i1}^{(\mathrm{QMAX2SAT})}$
$C_{i2}^{(\mathrm{QMAX2SAT})}$
$C_{i3}^{(\mathrm{QMAX2SAT})}$
$C_{i4}^{(\mathrm{QMAX2SAT})}$
Contradictory Information
Information Fusion Mechanism
Implementation into DHNN
Implementation into DHNN
Discrete Hopfield Neural Network (DHNN)
Local Field
Activation Function
Energy Function $H_{\mathrm{ZRAN2,3SAT}}$
Energy Function H_{QMAX2SAT}
Energy Function $H_{\mathrm{QMAX2,3SAT}}$
Energy Profile
-1 1 1 -1 1 -1 ... -1
Neuron States
Maximum Clause Satisfaction $\theta_{\max}$
Error Analysis
Ablation Analysis
Maximum Satisfaction Analysis
...